

Question Bank

Unit 2: Chemical Fuels

General & Calorific Value

1. Define fuel. Classify fuels with suitable examples.
2. What is calorific value of a fuel? Differentiate between Higher Calorific Value (HCV) and Lower Calorific Value (LCV).
3. Explain the principle of bomb calorimeter.
4. Write down the characteristics of an ideal fuel.
5. Discuss the importance of calorific value in selecting a fuel.

Solid Fuels (Coal)

6. What is proximate analysis of coal? Discuss the parameters measured.
7. What is ultimate analysis of coal? How does it differ from proximate analysis?
8. Compare proximate and ultimate analysis of coal.
9. What are the industrial applications of coal?
10. Explain the significance of carbon content in coal for its calorific value.

Liquid Fuels (Petroleum)

11. Explain the process of petroleum refining.
12. Discuss the process of petroleum cracking.
13. Differentiate between thermal cracking and catalytic cracking.
14. Define knocking in internal combustion (IC) engines. What are its ill effects?
15. What are anti-knocking agents? Explain with examples.
16. Discuss the difference between leaded petrol and unleaded petrol.
17. Write a short note on the environmental hazards of leaded petrol.

Gaseous Fuels (LPG & CNG)

18. What is LPG? State its composition and uses.
19. What is CNG? How is it different from LPG?
20. Discuss the advantages of using gaseous fuels over liquid and solid fuels.
21. Explain the role of CNG in reducing air pollution.
22. Write a short note on the storage and handling of LPG.

23. What are the applications of CNG in automobiles and industries?

Comparative & Conceptual

24. Compare solid, liquid, and gaseous fuels with examples.

25. Discuss the criteria for selecting a fuel for industrial applications.